Hooking It Up

The circuit shown in Figures 7-5 and 7-6 adds a pushbutton and trimpot, or other analog sensor, to our BlinkM to make an HSB Color Mixer. We have increased the complexity of our circuit a bit, but each of the components is fairly basic, using principles that you've seen before. We did have to make a slight modification to our pushbutton circuit by adding a small electrolytic capacitor across the positive and negative sides of the pushbutton. Electrolytic capacitors are polarized, so it is important that the white stripe attaches to the ground side of the switch. It's maybe not the best solution for hardware debouncing, but it will work for now, although other options can be found through a quick search online. While we had gotten away without using this before, the hardware interrupt happens so quickly that we need to debounce our switch using additional hardware to slow the switch down a little. This will prevent false readings without using a delay—which is necessary, as you will see in a moment.

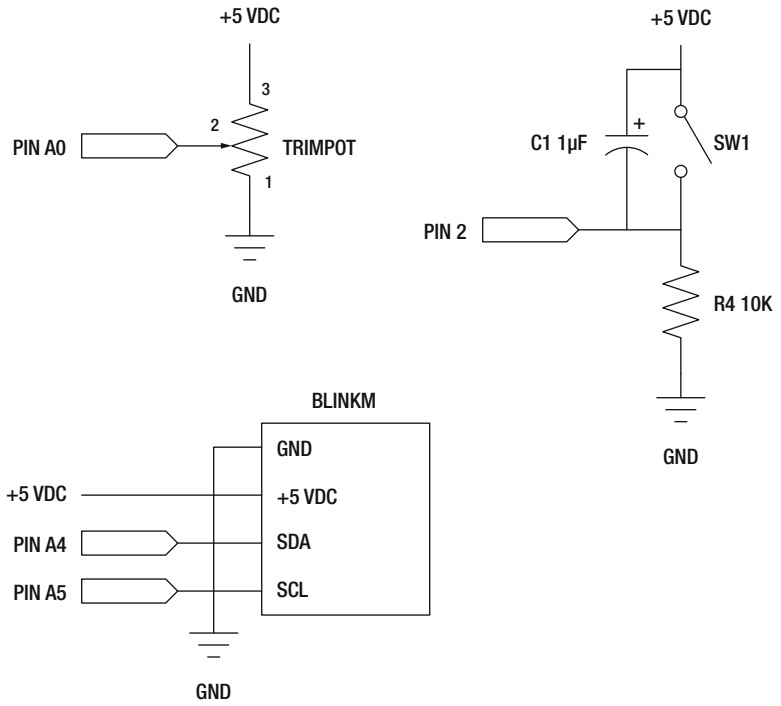

Figure 7-5. *HSB Color Mixer schematic*